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24k-0705. | BCS-4J.
Assignment #02.
Data-Base System

Question 1 -

a)
```
Select distinct c. customers name.
from customer. c.
where customer_id in(
        select p.customer_id.
        from purchases p.
        Join purchases_details pd on.
        On p.purchase_id =pd.purchase_id.
        Join Books b
        on pd.book_id =b.book_id.
        where b.price >(select avg(price)
                        from books 1,
```

b)
```
Select b1.title.
from books b1.
where b1.price >(
    select avg(b2.price)
    from book b2
    where b2.publicher=b1.publisher);
```

c)
```
Select c.customer_name.
from customer c.
where c.customer_id in
    (Select p.customer_id
     from purchases)
and. c.customer_id not in (
select p.customer_id
from purchases p.
join purchase_Details pd
on p.purchase_id=Pd.purchase_id.
join Books b on Pd.book_id=b.book_id.
where b.publisher = 'Pearson');
```

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d) Select b.title
From books b
where b.book_id not in
(Select Pd.book_id
from Pdpurchase_Detail);

e). Select a_autur_name.
from author a.
join Book_author
on a.author_id=ba.author_id.
join Books b on ba boolcid=b.book_id.
where b.price > (
Select avg(price)
from Boolcs);

f) Select P.customer_id, sum(Pd.quantity)
from Purchases P.
join Purchase_deteid.o
On P.purchase_id = Pd.Purchase_id.
Group P.customer_id.
Having sum(Pd.quantity) >10;

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Question 2:

a)
```
Select p.pessanger_name, b.bus_name, r.route_city, r.destinationcity
from passenger p
join Tickets t
on p.passenger_id = t.pasenger_id.
join Buses b
on t.buses_id = b.bus_id.
join routes r
on t.route_id = r.route_id;
```

b)
```
Select p.passenger_name, t.travel_date, t.fare.
From passenger p.
Join Tickets t
on p.passenger_id = t.passenge_id;
```

c)
```
Select b.bus_name, count (t.pessenger_id)
from buses b.
join Tickets t.
on b.bus_id = t.bus_id
group by b.bus_name;
```

d)
```
Select r.source_city, r.destination_city, p.passenger_name
from Tickets t.
join passenter p
on t.passenger_id = p.passenger_id.
join Routes r
on t.route_id = r.route_id.
```

e)
```
create or replace view PassengerTravel-summary as.
Select. p.passenger_name, b.buse_name, r.source_city,
        r.destination_city, t.travel_date.
join Tickets t
on p.passenger_id = t.passenger_id.
join Buses b
on t.bus_id = b.bus_id.
join Routes r
on t.route_id = r.route_id;
```

f) 1. Alter table Passengers
add contact_number varchar2(15);

2. Alter table tickets
modify fare Decimal(10,2);

3. Alter table Buses.
drop column capacity;

4. Drop table Routes cascade constraints.

Question 3.

a) → workout_programs.
(ProgramID, programName, Duration, Difficulty_level, coordinator_Name)

→ Trainers
(Trainerid, Trainer Name, contact Number)

→ Fitness Centers
(centerID, centerName)

b) → Program_Trainer (M:N (many-to-many))
→ Program-fitness Cntr (M:N (many-to-many))
→ Program - coordinator (1:1)

c) Trainer - Assignment (Resolving Program-trainer)
Attributes
(Program_id, Trainer_id, Role)
Primary key : (Program_id, trainer_id).

d) Coordinator_Name should not be modeled as a seperate entity becase.
- It does not have multiple Attributes.
- There is no additional information.
- Each program has exactly one coordinator, so no complex relationship exists.

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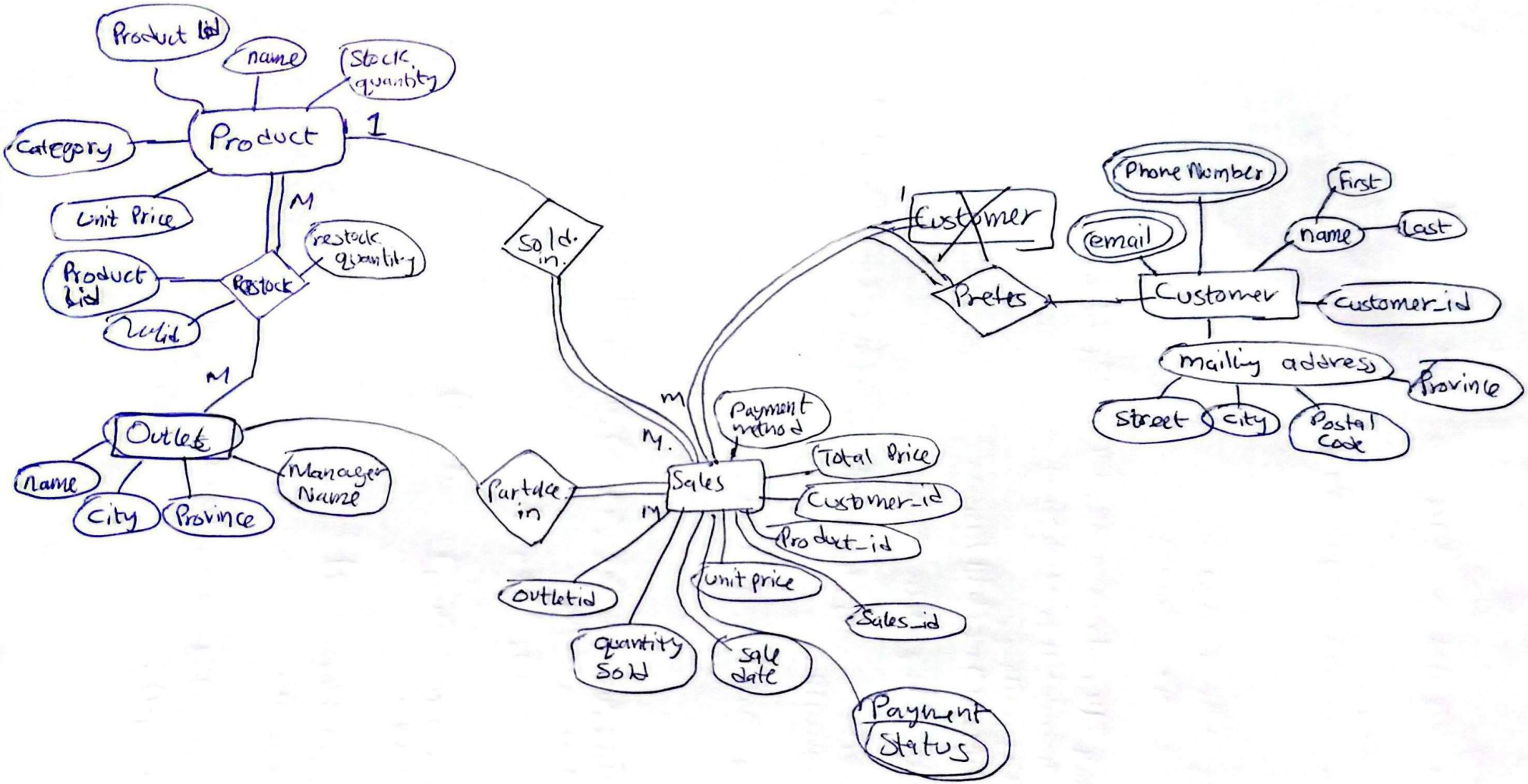
Question 4:


Product id
name
Stock quantity
Category
Product
1
Unit Price
M
restock quantity
Product id
Restock
Outlet id
M
Outlet
Name
City
Province
Manager Name
Sold in
Partake in
M
M
Payment method
Sales
Total Price
Customer_id
Product_id
Outletid
Unit price
Sales_id
quantity sold
sale date
Payment Status
Customer
1
Phone Number
First
email
name
Last
Prefes
Customer
customer_id
mailing address
Province
Street
City
Postal Code

Question 5:

1. Insertion Anomaly
   - Cannot insert a new AI Model/Agency/Platform without adding an article.

   Update Anomaly
   - Same data repeated so have to update on multiple places.

   Deletion Anomaly.
   - Deleting a record may remove important info.

2. Assumption
   - Each entity has a unique ID.
   - ID's determine their attributes.

FDs:
Article ID : Article Title, Submission date, Journalist ID, Platform Id.
JournalistID : JournalistName, JournalistEmail
PlatformID : PlatformName.
ATModelID : AIModelName, ModelVersion.

AgencyID : AgencyName, AgencyPhone.

VerificationID : Verificationdate, Verificationoutcome, ConfidenceScore, ArticleID, ATModelID, AgencyID.

3. 2NF (remove partial Dependencies)
Tables:

Articles:
(ArticleID, ArticleTitle, SubmissionDate, JourlistID, PlatformID)
P.K = ArticleID, ~~foregi~~
FK = JournalistID, PlatformID

Journalists:
(JournalistID, JournalistName, Journalistemail))
P.K = JournalistP.

Platforms
(PlatformID, PlatformName)
P.K PlatformID.

Agencies
AgencyID, Agency Name, Agency Phone)
PK : AgencyID.

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Verifications :

(VerificationID, ArticleID, AI ModelID, AgencyID, Verification Date, Verification outcome, confidence Score).

3NF (remove transitive dependencies)

- No transitive dependencies remain after seperation
- All non-key attributes depend only on P.K
- So tables remain the same as of 2NF

4. - Data Redundancy issues.
   - Agency Phone repeated. Causes update anomaly.
   - AI Model Name and version repeated. Causes inconsistency
   - Journalist Info repeated. Causes redundancy + update issue.

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